

Constrained Decoding for Gender-Fair French Language Generation in LLMs

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1. Introduction >>

- Gender bias long studied in NLP, lately focusing on biases stemming from interactions with LLMs^[1, 2]
- An understudied bias in gender-marked languages (e.g., French, German, Polish) is the **masculine generics (MG)** bias: **using the masculine form as a default/neutral form to refer to a mixed group of men/women**. Example in French:
Les **étudiants**^{MG} sont partis. (**Students**^{MG} left.)
- Psycholinguistic studies show **MG amplify male-centric mental representations**^[3–5]
- In generic conversational contexts, **LLMs also use MG**^[6, 7] and are **reluctant to use gender-fair language (GFL)**^[6]. Existing mitigations (e.g. gender rewriting^[8, 9]) operate only as **post-processing**, which fails to prevent the initial generation of biased text
- We propose a set of inference-based systems using constrained generation to promote GFL in French**

2. Methodology ⇄

- 3 constrained decoding systems** with 3 strategies a/b/c:
 - a. short:** force a gender-fair short suffix (e.g. ‘étudiant·es’)
 - b. coord:** force coordinated gendered pairs (e.g. ‘étudiants et étudiantes’) (forcing modes: (1) conjunction only, (2) conjunction + first token, (3) conjunction + full paired word)
 - c. neutral:** force a gender-neutral form (e.g. ‘corps étudiant’ [*student body*])

System	Mechanism	Strategies
hardforward	Forces an MG about to be generated into a gender-fair form	short, coord
hardbacktrack	Backtracks after an MG has been generated and replaces it with a gender-fair form	short, coord, neutral
hardexclusion	Bans MG tokens with NoBadWordsLogitsProcessor	—

3. Preliminary Findings 📄

- Tested with Llama 3.1 (8B), Ministral 3 (14B), EuroLLM (22B) & Qwen 3.5 (35B); Qwen 3.5 (2B) as context-verifier model
- Example with prompt “How much do students earn?” (*Combien les étudiants gagnent-ils ?*)

Vanilla (Llama 3.1)	Constrained (hardforward, short)
Le montant des allocations pour les étudiants en France peut varier en fonction de plusieurs facteurs [...] Cependant, les étudiants peuvent être éligibles à plusieurs types de bourses d’études [...]	Le montant des allocations pour les étudiant·es en France varie en fonction de plusieurs facteurs [...] Cependant, les étudiant·es peuvent bénéficier d’une aide financière appelée bourse d’études [...]

- Initial GFL forcing is sometimes enough to **nudge the model into generating GFL on its own** (i.e., when the constraint is applied only for the first 1/2 forcings and then fully removed)
- Performance benchmark on Llama 3.1 on H200, with 30 MG + 30 non-MG prompts, mean of 10 runs per condition:

Condition	Overhead (×)	ms/tok
Vanilla	1.00	8.9
GFL LogitProc. (hf)	1.04	9.3
Post-Gen Rewrite	2.02	18.0
GFL LogitProc. + Verifier (hf) [†]	2.07	18.4
Post-Gen Rewrite (long sysprompt)	2.07	18.4

[†] Between 5 (MG) and 2 (non-MG) mean verifier calls.

4. Prospects 🚀

- Evaluation via a three-dimensional Pareto analysis:

$$\hat{M} = \frac{MG}{HN} \quad \hat{I} = \frac{I}{HN} \quad \hat{N} = \frac{N}{HN}$$

- where HN = human noun, *I* = inclusive form, *N* = neutral form; minimizing $\hat{M}$ and maximizing $\hat{I}$, $\hat{N}$
- Comparison with vanilla models (coherent generation, instruction-following) and with models fine-tuned on gender-fair instruction responses

References

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